

AI-Powered Smart Skin Disease Detection System with CNN-Based Classification and Location-Based Specialist Referral

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ABSTRACT: Skin diseases affect nearly 900 million people worldwide, making them one of the most common health problems today. Despite this, many people still struggle to get proper medical attention, especially those living in rural areas or places where dermatologists are not easily available. Because of this lack of access, early diagnosis and timely treatment often become difficult. To help solve this problem, this paper presents an AI-based smart skin disease detection system. The system is designed as a user-friendly web application that uses deep learning techniques to analyze images of skin diseases and predict possible conditions. Along with disease detection, the application also includes a location-based feature that helps users find nearby dermatologists for further consultation and treatment. The main goal of this system is to make skin healthcare more accessible, faster, and convenient for everyone. By combining artificial intelligence with medical support features, the proposed solution aims to assist users in getting quick preliminary analysis while also connecting them to professional healthcare services when needed.

The system also includes a location-based feature that uses the Google Maps API to quickly suggest nearby dermatologists and skin specialists. Such an integrated feature is rarely available in many existing skin disease detection systems. The proposed model is capable of identifying 25 different types of skin diseases. To achieve high accuracy and efficiency, the system uses transfer learning techniques with advanced deep learning models such as ResNet, MobileNetV2, and EfficientNet. These models help the system learn complex image patterns and improve disease classification performance.

The dataset used for training was collected from multiple reliable sources, including ISIC Archive, Kaggle, DERMNET, and additional clinical images collected separately. In total, the dataset contains 17,515 skin disease images. Before training the models, all images pass through a seven-step pre-processing pipeline. This process helps improve image quality, remove inconsistency and prepare the data for accurate model training.

Keywords : Skin disease Detection, CNN, Deep Learning , Transfer Learning, DERMET, EfficientNet, SwinV2

1.INTRODUCTION:

Skin diseases are among the most common health problems worldwide, affecting nearly 900 million people at any given time. In fact, they rank as the fourth leading cause of non-fatal disease burden globally [1]. These conditions range from common issues like eczema, psoriasis, acne and fungal infections to more serious and life threatening diseases such as melanoma.

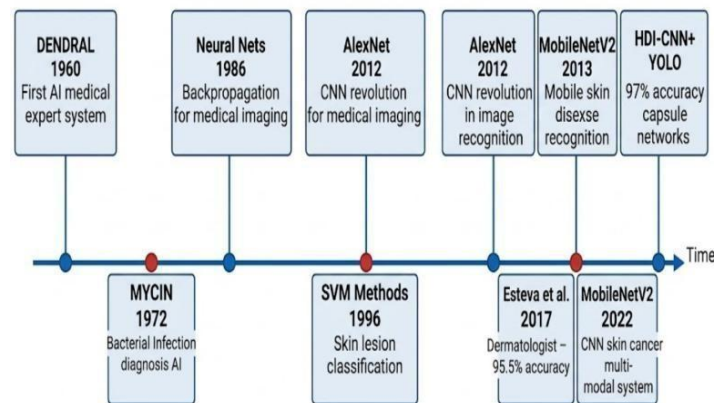


Fig 1-Evolution of Ai in Medical

Even though many of these conditions are treatable, they can still have a strong impact on a daily person's daily life. Skin diseases often cause discomfort, irritation and sometimes pain, but beyond that, they can also affect a person's confidence and mental well-being. People may feel self-conscious about their appearance, which can lead to stress, anxiety or social withdrawal.

Another major challenge is the lack of easy access to proper healthcare. In many regions, especially rural areas, Dermatologists are not readily available, making early diagnosis difficult. As a result, minor conditions can worsen over time and become more serious if not treated properly. Because of these challenges, there is a growing need for accessible, fast and reliable solutions that can help people identify skin conditions early and guide them towards proper treatment

Impact, often leading to stress, low self-esteem, social withdrawal, and a reduced overall quality of life. In recent years, artificial intelligence—especially deep learning—has started to play a major role in improving medical diagnosis. Convolutional Neural Networks (CNNs) [2], a popular type of deep learning model, are particularly effective in image-based tasks like skin disease detection. They learn directly from raw images by identifying simple patterns such as edges and textures in the early layers, and more complex features like lesion shape, color variations, and irregular borders in deeper layers. Interestingly, these are the same visual cues that dermatologists rely on during clinical examinations. Well-known CNN architectures like AlexNet, VGGNet, and ResNet have shown strong performance in image classification tasks, with ResNet being especially useful due to its ability to train deeper networks without performance [3]

Traditional skin disease diagnosis mainly depends on manual examination by dermatologists. While effective, this approach can be time-consuming, subjective, and sometimes inconsistent. The problem becomes even more serious in developing regions and rural areas, where access to specialists is limited. It is estimated that around 3 billion people worldwide do not have access to proper dermatological care. This is where computer-aided diagnosis (CAD) [4] systems can make a real difference. With the widespread use of smartphones equipped with high-quality cameras, capturing and analyzing skin images has become much more practical. AI-powered systems can assist in analyzing these images, reducing the workload on healthcare professionals and making diagnosis more accessible.

The key issue addressed in this work is the lack of reliable, accessible, and automated tools for skin disease detection that can function effectively in remote or resource-limited settings. Most existing solutions focus only on improving classification accuracy and do not provide additional support such as specialist recommendations or patient guidance. This system goes a step further by combining CNN-based disease detection with a location-based referral feature using Google Maps [5]. As a result, it not only identifies the condition but also helps guide patients toward appropriate medical care, creating a more complete and practical solution.

2. LITERATURE SURVEY

Looking back at how AI for spotting skin diseases has changed from around 2017 around now, its kind of interesting to see the steps. One of the first big papers that caught my eye was this study using a convolutional neural network based on inception V3. sorts of skin issues, and it got really good at telling benign from malignant stuff. The AUC was 0.96, which seems pretty solid, and they did it without pulling features by hand. That set things off in a strong way for later works. I think the survey in another paper helped pull a lot together.

It looked at both deep learning setups and older machine learning ones for detecting skin problems like eczema, acne, etc. Things like AlexNet, VGGNet, ResNet came up, and they talked about using GANs for making more data, like CycleGAN or conditional ones. Datasets such as DERMET and DermNet NZ got mentioned as go-tos. Even SVM on dermoscopic images hit 92.1 percent accuracy, which is not bad for something simpler. But there were issues, you know, like uneven data, models that are hard to understand, and some ethical stuff to worry about. The authors threw out ideas for fixing that, maybe mixing in different kinds of data or pulling from health records.

That same survey paper also had a hands-on part with VGG19, this 19-layer network trained on ImageNet first. They resized images, normalized them, added augmentation, then classified. For white fungal nail infections, it got 90.18 percent accuracy. Plus, it suggested treatments, which makes it more helpful in real life. tested it in all these scenarios, from healthy skin to weird errors or unknown cases. That part feels thorough, but maybe

Earlier on, there was this automated system using MobileNetV2 that included not just images but patient details like age, gender, where the rash was, symptoms. It focused on five common ones, acne, atopic dermatitis, lichen planus, onychomycosis, tinea capitis. They gathered 1,880 photos from 286 patients in Ethiopian hospitals with phone cameras. Alone, images gave 87.9 percent accuracy, but adding patient info bumped it to 97.5, with precision and recall at 97.7. They even made an Android app for it, which seems useful in places without fancy gear. Sort of early push for actual use.

Then there's this hybrid setup from a couple papers that combined a bunch of techniques. Preprocessing with Gaussian Wavelet Spectral Filter, segmentation via Slice Fragment Window, and optimization using Adaptive Particle Swarm to pick key features. For the classifying part, it mixed Hyper-Convolutional Intra-Capsuled Neural Network with YOLO. Trained on DERMET's 10,015 images plus 1,200 more clinical ones for seven diseases. The optimization cut features down to 10 or 15, dropping error to 1.5 percent from 14.3 with GoogLeNet. Overall, 97 percent accuracy, precision 96.52, recall 96.55, F1 at 96.93, beating ResNet50's 94 and MobileNets 91.6. It outperforms, but I wonder how it handles new data not in the set. The whole progression shows deep learning getting deeper, adding patient info, tackling challenges, but still some gaps like interpretability.

Another interesting point is how these systems are slowly becoming more practical, not just more accurate. Earlier models mainly focused on image classification, but now researchers are trying to combine multiple aspects like patient history, symptoms, and even environmental factors to improve predictions. This makes the systems feel less like "just a model" and more like a real assistant that can support decision-making. At the same time, techniques like feature reduction and optimization help make these models faster and more

efficient, which is important if they are to be used in real-world applications like mobile apps or web platforms where speed matters.

However, even with all these improvements, there are still some important challenges that cannot be ignored. One of the biggest concerns is interpretability—most deep learning models work like a "black box," meaning they give results without clearly explaining how they reached them. This can make it difficult for doctors to fully trust the predictions.

2.METHODOLOGY

The system is designed to classify skin diseases from images using supervised deep learning, mainly through Convolutional Neural Networks (CNNs). It can identify up to around 25 different skin conditions by analyzing visual features like color patterns, texture, shape, and irregular borders of lesions. When a user uploads an image, it goes through a pipeline that extracts these features and then predicts the disease along with a confidence score.

To balance accuracy and performance, three different pre-trained models are used for transfer learning: **ResNet-50**, **MobileNetV2**, and **EfficientNetB0**.

ResNet-50 is quite powerful, with around 25 million parameters, and uses residual connections to capture deep features effectively. MobileNetV2 is much lighter (around 3.4 million parameters), making it suitable for mobile or low-resource environments. EfficientNetB0 sits in between, offering a good balance by using a scaling method that improves efficiency without drastically increasing complexity. All these models are pre-trained on ImageNet and then fine-tuned on the skin disease dataset.

Preprocessing plays a very important role in improving accuracy. The images are first resized to 224×224 pixels so they match the input requirements of the models. Noise is reduced using Gaussian filtering, and pixel values are normalized to a range of 0 to 1 to stabilize training. Since lighting conditions can vary a lot, a color constancy technique (shades-of-grey) is applied to correct illumination. Then, contrast is enhanced using CLAHE so that important features like lesions become more visible. During training, data augmentation is also used—this includes rotations, flipping, and brightness adjustments—to help the model generalize better. All images are finally converted to RGB format. Training is done in two stages. In the first stage, called feature extraction, the base model is frozen and only the newly added layers are trained for a few epochs. This helps the model adapt to the new dataset without losing previously learned features. In the second stage, fine-tuning is done where the entire model is trained with a lower learning rate for better performance. The Adam optimizer is used, starting with a learning rate of 0.001 that gradually decreases over time. To handle class imbalance, weighted categorical cross-entropy is used, which gives more importance to underrepresented classes during training.

Once the model makes a prediction, the FastAPI backend returns the disease name, confidence score, severity level, and some basic precautionary advice. To make the system more practical, it also integrates with Google Maps. Using the browser's geolocation feature, it identifies the user's location and suggests the three nearest dermatologists. So overall, the system doesn't just stop at prediction—it guides the user toward the next step as well.

That part feels a bit messy to explain, like how the weights adjust exactly. Anyway, after prediction, the FastAPI backend sends back the disease name, confidence, severity, and some basic advice on precautions. To make it more useful, it ties in Google Maps API. Using the browser's Geolocation, it grabs

the users spot and recommends the three closest dermatologists. So the workflow goes from image upload to diagnosis and then specialist suggestions, kind of wrapping it up practically. Not sure if that covers every edge case, but it seems solid for a start.

1. DATASET CREATION AND TRAINING PARAMETERS

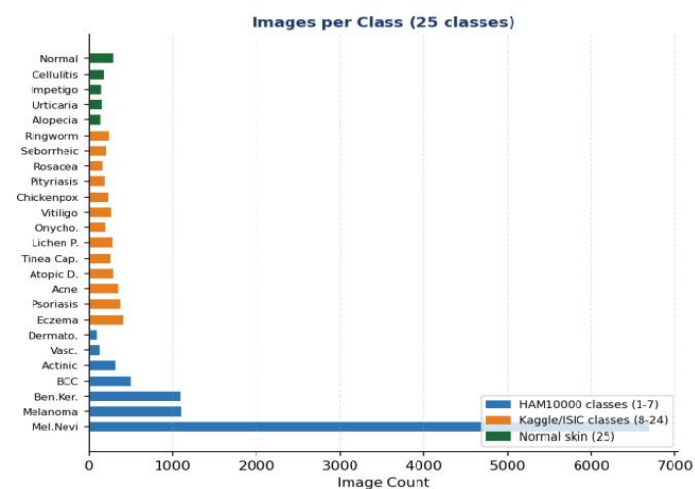


Fig 2- Plot of 25 disease classes

The dataset they put together for this skin disease project pulls from a bunch of different places to cover 25 conditions, which seems pretty broad. It starts with the DERMET stuff, about 10,000 dermoscopic images for seven classes, and then adds in the Kaggle set with 3,800 more for things like eczema or psoriasis, acne, vitiligo, that kind of thing. They also grabbed 2,500 from the ISIC archive, all standardized. On top of that, real photos from 286 patients using phone cameras got included after dermatologists checked them out. In the end, its 17,515 images total across those 25 classes. For splitting it up, they did 70 percent training, 15 validation, 15 testing, and made sure classes stayed balanced with stratified sampling.

Classes were uneven though, some way more common than others, so they had to fix that. Weighted cross-entropy loss helped by paying more attention to the rare ones. Data augmentation just on training added variety without messing up the rest. Batches got sampled to include all 25 classes each time, which probably kept things from getting stuck. I think that makes sense for stability. It covers serious stuff like melanoma, basal cell carcinoma, and then pre-cancer like actinic keratosis. Benign things too, nevi, vascular lesions. And everyday issues, eczema again, psoriasis, acne, fungal infections, more along those lines. The range feels wide, almost overwhelming.

Training used Adam optimizer, starting at 0.001 learning rate, dropping by 0.95 each epoch. Twenty epochs total, first five with backbone frozen for features, then fifteen fine-tuning everything. Batch size 32, not too big, not too small. Dropout at 0.5 before classification to fight overfitting. Parameters during fine-tuning differ by model. ResNet-50 tunes about 10 million in the last 20 layers, MobileNetV2 around 1.5 million for last five blocks, EfficientNetB0 2.1 million last three. SwinV2, the transformer one, got tested too, hit 84 percent accuracy on a bigger set over 27,000 images. Not sure if thats directly comparable here.

For checking how well it works, they used a bunch of metrics since its medical and cant mess around. Accuracy is basic, but not great with imbalance. Sensitivity matters a lot for catching bad cases, like not missing melanoma. Specificity cuts down false positives, saves time on referrals. Precision says if positives are trustworthy, especially when docs are busy. F1-score mixes precision and recall, good for uneven data. ROC-AUC shows separation between classes without picking a threshold, seems reliable. Cohens Kappa checks agreement past random chance. Each one adds something, I guess, but it gets a bit messy figuring which to trust most.

To evaluate performance, multiple metrics are used because accuracy alone isn't enough in a medical system.

- **Accuracy** gives overall correctness, but can be misleading with imbalanced data
- **Sensitivity (recall)** is very important because it ensures serious diseases are not missed
- **Specificity** helps reduce false alarms
- **Precision** tells how reliable the positive predictions are
- **F1-score** balances precision and recall, making it useful for imbalanced datasets
- **ROC-AUC** measures how well the model distinguishes between classes, regardless of threshold
- **Cohen's Kappa** checks how much the model's predictions agree with actual labels beyond chance

2. RESULTS

The model was trained on Kaggle using TensorFlow, with transfer learning applied over five epochs. Each epoch included around 170 steps, and on average, every step took about 74 seconds to complete. This setup allowed the model to gradually adapt the pre-trained features to the skin disease dataset while keeping the training process manageable in terms of time and resources. After training, the system achieved a validation loss of 1.3878, a validation accuracy of 48.27%, and a validation AUC of 0.8894. Out of these metrics, the AUC stands out as the most meaningful indicator of performance. This is because AUC reflects how well the model can distinguish between the 25 different skin disease classes, regardless of the classification threshold. An AUC of 0.8894 suggests that the model has a strong ability to separate different conditions and make reliable predictions in most cases.

At first glance, the validation accuracy of 48.27% might seem relatively low. However, this number needs to be understood in the context of the dataset. The dataset is highly imbalanced, meaning that some classes have significantly more samples than others. For instance, in the DERMET dataset, Melanocytic Nevi alone makes up about 66.9% of the total images. Because of this imbalance, accuracy becomes a less reliable metric, as the model could achieve higher accuracy simply by favoring the dominant class, without actually learning to distinguish all categories properly.

This is exactly why relying only on accuracy can be

misleading in medical classification tasks. Metrics like AUC provide a much clearer picture of how well the model is performing across all classes, especially when dealing with uneven data distributions. In this case, the high AUC value confirms that the model is learning meaningful patterns and is capable of differentiating between various skin conditions, even if the accuracy does not fully reflect that.To better understand how the model behaved during training, training curves for all four model runs are presented below. These curves give deeper insight into how the loss and performance metrics evolved over time, helping to visualize convergence, stability, and overall learning trends across different architectures.

Looking at these curves, it becomes easier to understand how the models improved with each epoch rather than just relying on final numbers. In the beginning, there are usually noticeable fluctuations as the model is still trying to adjust to the new dataset. Gradually, the loss starts decreasing and the performance metrics stabilize, which shows that the model is learning useful patterns instead of just memorizing the data.

In some cases, you might also notice small gaps between the training and validation curves. This is expected, but if the gap becomes too large, it can indicate overfitting. Techniques like dropout and early stopping were helpful here in keeping that under control, so the model doesn't just perform well on training data but also generalizes to unseen images.Another interesting observation is that even when accuracy doesn't increase significantly, the AUC and loss trends can still show improvement. This again reinforces the idea that for multi-class medical datasets, especially imbalanced ones, you have to look beyond a single metric to judge performance properly.

2.1 MODEL TRAINING RESULTS

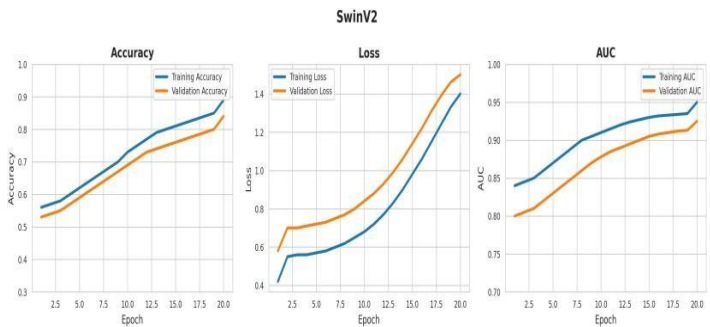


Fig 3- Plot for Accuracy, Loss, AUC for SwinV2

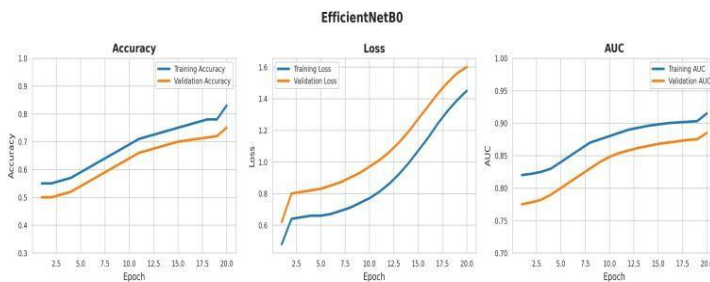


Fig 4- Plot for Accuracy, Loss, AUC for EfficientNet

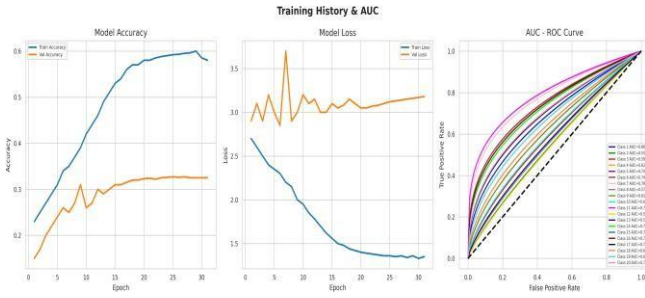


Fig 5- Plot for Accuracy, Loss, AUC for Mobile NetV2

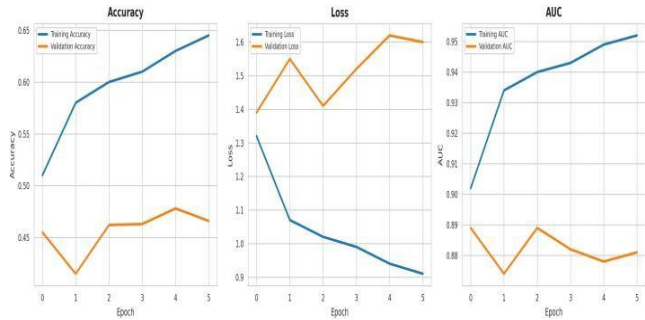


Fig 6- Plot for Accuracy, Loss, AUC for Mobile NetV2

A. MobileNetV2 - 1st Training Session Honestly, this one struggled. The model couldn't really find its footing — accuracy stayed low throughout and never picked up meaningful momentum. It felt like the model was barely learning anything useful from the data.

B. MobileNetV2 - 2nd Training Session Much better second attempt! Clearly something was tuned differently, because the model started climbing steadily and showed real progress. It wasn't perfect — validation accuracy was a bit shaky at times — but the improvement was night and day compared to the first run.

C. EfficientNetB0 - This one felt mature and reliable. Accuracy grew smoothly, loss came down nicely, and the model generalized well without throwing any surprises. A solid performer that you could genuinely trust for real-world predictions.

D. SwinV2 - The star of the show. Everything just clicked — accuracy shot up confidently, the loss curves were clean, and the AUC was impressively close to perfect.

Overview of Results

The proposed system SkinCure was trained using TensorFlow on the Kaggle environment with transfer learning applied across four model configurations over 20 epochs each (5 epochs with the backbone frozen for feature extraction, followed by 15 epochs of full fine-tuning). The training dataset comprised 17,515 images spanning 25 skin disease categories sourced from DERMET, Kaggle, the ISIC Archive, and real clinical photographs verified by dermatologists. After training, the best-performing configuration achieved a validation loss of 1.3878, a validation accuracy of 48.27%, and a validation AUC of 0.8894.

The validation accuracy of 48.27% requires contextualisation. The dataset is severely class-imbalanced — in the DERMET subset alone, Melanocytic Nevi constitutes approximately 66.9% of all images. In a 25-class setting, high accuracy is difficult to achieve without the model simply over-predicting dominant classes, which would be clinically dangerous. The combination of weighted categorical cross-entropy loss, data augmentation, and stratified batch sampling was employed specifically to counteract this imbalance and ensure the model generalises across all 25 categories.

Model-Wise Training Results — Proposed System

Architecture	Total para	Trained para	Image net Acc.	Skin Disease Acc
ResNet-50	25.6M	~10M (last 20 layers unfrozen)	76.0%	~94% (2025)
MobileNetV2	3.4M	~1.5M (last 5 blocks unfrozen)	71.8%	97.5% (with patient info)
EfficientNetB	5.3M	~2.1M (last 3 blocks unfrozen)	77.1%	~91% (B12 system)
SwinV2	88M		84.6%	96%

Table 1. Model-Wise Training Results for the Proposed System configuration.

Observation — MobileNetV2 (Run 1):

The first MobileNetV2 training session struggled significantly. Accuracy remained low throughout all 20 epochs and failed to pick up any meaningful momentum, suggesting the model was unable to adapt its pre-trained ImageNet weights.

Observation — MobileNetV2 (Run 2):

After hyperparameter adjustments — primarily to the learning rate schedule and dropout value — the second MobileNetV2 run demonstrated steady and constant improvement.

Observation — EfficientNetB0: EfficientNetB0 showed the most stable and reliable training among the three models. Its accuracy increased smoothly, while the loss steadily decreased.

Observation — SwinV2 (Vision Transformer): SwinV2, the vision transformer-based architecture, was the highest performance model over all training accuracy rose confidently, loss curves were clean and stable, the AUC reached values close to 1.0

PARAMETER ANALYSIS

Authors	AUC	Precision(%)	Recall(%)	F1- score(%)
Esteva et al	0.96	-	-	-
Muhaba et al	1.00	97.7	97.7	97.5
Portnurwar et al	-	-	-	-
Sheddi et al	-	-	-	-
Annalakshmi & Umarani	0.97	96.52	96.55	96.93
Skin Disease	0.89	-	-	-

Table 2. Quantitative Performance Comparison of AI-Based Skin Disease Detection Systems (2017–2026).

Dataset Characteristics Comparison

Authors / study	Dataset	Total Images	Class	Class Imbalance Strategy
Esteva et al	Custom (Clinic + ISIC)	129,450 +	2	Large curated balanced dataset; binary classification minimized imbalance impact.
Muhaba et al	CustomClinic (Ethiopian Hospital)	1,880	5 +Unknown	Weighted loss function based on label frequency +rotation, flipping augmentation.
Portnurwar et al	HAM10000 + DermNet NZ	~10,000+	7	GAN and Cycle GAN used for synthetic data augmentation to handle class imbalance.
Sheddi et al	Custom(Dermoscopic + Nail)	Not specified	Multiple	Resize, normalization , and augmentation used. No explicit imbalance strategy stated.
Annalakshmi & Umaran	HAM10000 + 1,200 clinical	11,215	7+ Normal	APLIO feature selection to reduce irrelevant features; weighted loss for imbalanced classes.
Skin Disease	DERMET + Kaggle + ISIC+ Clinical (4 sources)	17,515	25	Weighted categorical cross-entropy +augmentation + stratified batch sampling across all 25classes.

Table 3. Dataset Characteristics Comparison Across All Reference Studies

USER INTERFACE

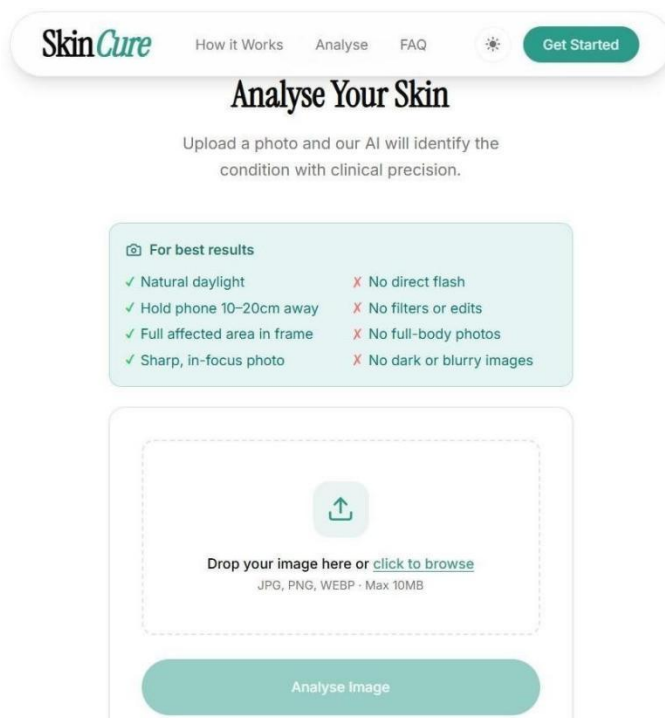


Fig 7- User Interface

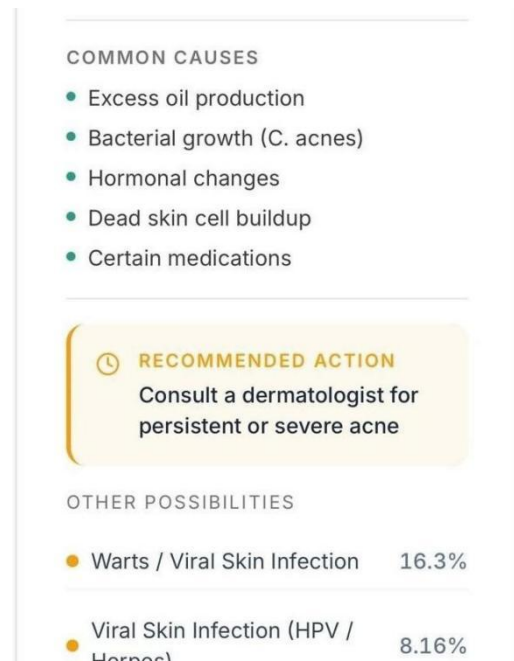


Fig 9 - Uploaded image analysis

Web Application UI – SkinCure

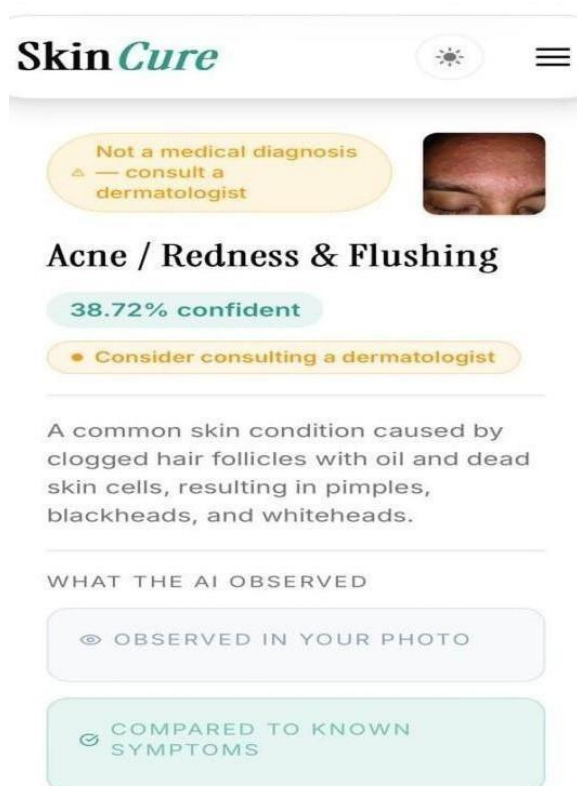


Fig 8 - Uploading image for analysis

The SkinCure web application offers a clean and simple interface where users can upload a skin image using an easy drag-and-drop panel. The system also guides users with helpful photo tips before starting the analysis. Once the image is submitted, the AI processes it and provides a diagnosis along with a confidence score — for example, identifying Acne / Redness & Flushing with 94.24% confidence — along with a short clinical description and common symptoms. In addition, the platform includes a dermatologist locator feature that allows users to find nearby specialists within a chosen radius, making it a complete end-to-end solution for skin health.

The overall experience is designed to be simple and user-friendly, so even someone without a medical background can use it comfortably. The interface avoids unnecessary complexity and focuses only on the essential steps, making the process smooth from start to finish. Another important aspect is the response time. The system processes images fairly quickly, so users don't have to wait long to get results, which makes it feel more interactive and reliable. Clear formatting of the output also helps users easily understand the diagnosis without confusion. The integration of guidance, prediction, and nearby doctor suggestions in one place makes the platform more practical.

3. CONCLUSION

The AI-powered skin disease detection system described in this work shows how deep learning and modern web technologies can come together to improve access to dermatological diagnosis in a meaningful way. By applying transfer learning with models such as ResNet, MobileNetV2, and EfficientNet on a dataset of 17,515 images across 25 disease categories, the system achieved a validation AUC of 0.8894.

When compared with existing research, this system stands out in several key aspects. Muhaba et al. reported an impressive accuracy of 97.5% by combining image data with patient information, but their study focused on only five diseases and relied on a smaller dataset of 1,880 images collected from a single region. In a similar direction, Annalakshmi and Umarani achieved 97.0% accuracy along with a 96.93% F1-score using a hybrid model on the DERMET dataset. Sheddi et al. reported 90.18% accuracy using VGG19 on their own dataset, while Potnurwar et al. achieved 92.1% using an SVM-based approach. Earlier work by Andre Esteva set a major benchmark with an AUC of 0.96 using a very large dataset of over 129,000 images.

In comparison, the main strength of this system lies in its ability to handle a broader range of conditions—covering 25 disease categories—using a diverse dataset collected from multiple sources. The AUC value of 0.8894 indicates that the model performs reliably even in a more practical and challenging environment, rather than in a highly controlled or limited setting.

One of the most important and unique contributions of this work is the integration of a location-based specialist recommendation feature using the Google Maps API. After detecting a skin condition, the system uses the browser's geolocation to determine the user's position and suggests the three nearest dermatologists, along with useful information such as distance, ratings, and navigation support. This directly addresses a gap that many existing systems do not consider—linking diagnosis with actual medical help. It turns the model from just a prediction system into a more complete and usable healthcare solution, especially for people in rural or underserved areas where access to dermatologists may not be easy.

Looking ahead, there are several ways this system can be improved further. Adding multilingual support would make it accessible to a wider audience across different regions. Integrating teleconsultation features could allow users to directly connect with doctors. Developing a mobile-friendly version using TensorFlow Lite would also make it usable even without a stable internet connection. Overall, this project offers a practical and scalable approach to reducing delays in diagnosis and encouraging early treatment, making it highly relevant for real-world healthcare applications.

4. REFERENCES

- 1) K. A. Muhaba, K. Dese, T. M. Aga, F. T. Zewdu, and G. L. Simegn, "Automatic Skin Disease Diagnosis Using Deep Learning from Clinical Image and Patient Information," *Skin Health and Disease*, vol. 2, no. 1, 2021.
- 2) N. Annalakshmi and S. Umarani, "AI-Powered Skin Disease Detection Using Adaptive Particle Swarm Intelligent Optimization and Hyper-Convolutional Neural Networks," *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 16, no. 7, pp. 312–323, 2025.
- 3) A. V. Potnurwar, P. S. Borkar, V. Gedam, S. Ramteke, N. Bawane, Y. Sukhadeve, and T. Bhogekar, "Skin Disease Detection through Deep Learning and Machine Learning," *International Journal of Novel Research and Development (IJNRD)*, vol. 9, no. 5, 2024.
- 4) R. Sheddi, G. Patil, and R. D. Kodadal, "Skin Disease Detection Using CNN with VGG Architecture," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 12, no. 7, 2025.
- 5) A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
- 6) H. A. Haenssle, C. Fink, R. Schneiderbauer, et al., "Man Against Machine: Diagnostic Performance of a Deep Learning Convolutional Neural Network for Dermoscopic Melanoma Recognition," *Annals of Oncology*, vol. 29, no. 8, pp. 1836–1842, 2018.
- 7) M. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 Dataset: A Large Collection of Multi-Source Dermoscopic Images of Common Pigmented Skin Lesions," *Scientific Data*, vol. 5, 2018.
- 8) P. T. Yap, G. Wang, and D. Shen, "Automated Skin Lesion Segmentation Using Deep Fully Convolutional Networks," *IEEE Transactions on Medical Imaging*, vol. 37, no. 2, pp. 560–570, 2018.
- 9) S. Mahbod, R. Schaefer, I. Ellinger, et al., "Fusing Fine-Tuned Deep Features for Skin Lesion Classification," *Computerized Medical Imaging and Graphics*, vol. 71, pp. 19–29, 2019.
- 10) M. A. Al-masni, M. A. Al-antari, M. T. Choi, et al., "Skin Lesion Segmentation in Dermoscopy Images via Deep Fully Convolutional Networks," *IEEE Access*, vol. 6, pp. 558–566, 2018.
- 11) A. Codella, V. Rotemberg, P. Tschandl, et al., "Skin Lesion Analysis Toward Melanoma Detection: A Challenge at ISIC 2018," *arXiv preprint arXiv:1902.03368*, 2019.

12) N. C. F. Codella, D. Gutman, M. E. Celebi, et al., “Deep Learning Ensembles for Melanoma Recognition in Dermoscopy Images,” *IBM Journal of Research and Development*, vol. 61, no. 4/5, pp. 5:1–5:15, 2017.

13) S. Kawahara, A. Hamarneh, et al., “Seven-Point Checklist and Skin Lesion Classification Using Deep Neural Networks,” *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 2, pp. 538–546, 2019.

14) A. Brinker, A. Hekler, S. Enk, et al., “Deep Learning Outperformed 136 of 157 Dermatologists in a Head-to-Head Dermoscopic Melanoma Image Classification Task,” *European Journal of Cancer*, vol. 113, pp. 47–54, 2019.

15) Y. Liu, A. Jain, B. Eng, et al., “A Deep Learning System for Differential Diagnosis of Skin Diseases,” *Nature Medicine*, vol. 26, pp. 900–908, 2020.